AI COMMUNICATION SYSTEMS STUDY KIT

Presentation

Amplitude Modulation (AM) and
Frequency Modulation (FM)

Prepared for an ECE learning project

1. Introduction

Communication system: A system that transfers information from a source to a destination.

Modulation: The process of varying a carrier signal according to the information or message signal.

Modulation helps transmit information over communication channels.

2. Project Objectives

Understand the basic idea of AM and FM.

Generate and observe message, carrier, AM, and FM waveforms.

Compare amplitude and frequency changes in the waveforms.

Provide a simple learning resource with explanations, graphs, glossary, quiz, audio, and video.

3. Basic Communication Model

Message signal: The original information to be transmitted.

Carrier signal: A high-frequency signal used to carry the message.

Modulator: Combines the message and carrier by varying a carrier property.

Channel and receiver: Carry the signal and recover the information at the destination.

4. Amplitude Modulation (AM)

In AM, the amplitude of the carrier changes according to the message signal.

The carrier frequency remains constant in the standard AM model.

The outer envelope of the AM waveform follows the message signal.

Equation: $s(t) = A_c[1 + \mu \cos(\omega_m t)] \cos(\omega_c t)$

Here, A_c is carrier amplitude, μ is modulation index, ω_m is message angular frequency, and ω_c is carrier angular frequency.

5. Frequency Modulation (FM)

In FM, the instantaneous frequency of the carrier changes according to the message signal.

For ideal FM, the carrier amplitude remains constant.

The spacing of the oscillations changes as the instantaneous frequency varies.

Equation: $s(t) = A_c \cos[\omega_c t + \beta \sin(\omega_m t)]$

Here, β is the FM modulation index for a sinusoidal message.

6. AM and FM Comparison

AM: Information is represented by changes in carrier amplitude.

FM: Information is represented by changes in instantaneous carrier frequency.

Noise: FM generally resists amplitude noise better than AM when used with suitable receivers.

Applications: AM is used in some broadcast radio services; FM is widely used in broadcast radio and other communication systems.

7. Simulation Workflow

1. Choose message frequency, carrier frequency, sampling frequency, and simulation duration.

2. Generate the message signal and carrier signal.

3. Create the AM signal using the selected modulation index.

4. Create the FM signal using the selected frequency deviation or modulation index.

5. Plot and compare the waveforms.

6. Save the graphs and explain the observations.

8. Observations from the Graphs

The message signal is the information waveform.

The carrier is a higher-frequency oscillation.

In AM, the envelope varies with the message signal.

In FM, the waveform's instantaneous oscillation rate changes while ideal amplitude remains constant.

Use the actual graphs generated by your code in this section of the final report.

Note: This slide explains expected waveform features; it does not claim that the simulation code was independently audited.

9. AI Learning-Kit Features

A beginner-friendly explanation of communication concepts.

Simulation graphs to support visual learning.

A glossary of key technical terms.

A quiz with answers for self-assessment.

An English audio narration and narrated slide video as optional learning resources.

10. Conclusion

AM and FM are two important analog modulation techniques.

AM changes carrier amplitude; FM changes instantaneous carrier frequency.

Waveform simulations make the differences easier to understand.

The Study Kit brings together explanations, graphs, glossary, quiz, and media resources in one learning project.

11. Viva Questions

Q1. What is modulation?
Answer: Varying a carrier property according to the message signal.

Q2. What changes in AM?
Answer: Carrier amplitude.

Q3. What changes in FM?
Answer: Instantaneous carrier frequency.

Q4. What does the AM envelope represent?
Answer: The message signal, for standard AM under suitable conditions.

Q5. What is the purpose of the project?
Answer: To help learners understand AM/FM through explanations and simulations.

12. Submission Checklist

Include the project introduction and objectives.

Attach the actual AM/FM graphs generated by your simulation.

Include source code or a link to the code repository.

Add glossary, quiz, and answer key.

Add the audio and video links or QR codes if required.

Verify all links open correctly before submitting.

Replace any generic or placeholder information with your actual name, department, college, code link, and simulation results before submission.